

AI Learning Assistant Pro

Session Report • May 03, 2026

■ Summary

Java is a programming language used to create computer programs, apps, and websites. It's like writing a set of instructions that a computer can understand to perform specific tasks.

■ Conversation

Chat: Java

Q1:

java

A1:

■ **Java Programming** Java is a high-level, object-oriented programming language that is widely used for developing a vast range of applications, including Android apps, web applications, desktop applications, and enterprise software. --- ■ **Java Basics** * **Variables** → A variable is a name given to a location in memory where a value can be stored. In Java, you can declare variables using the ``data_type variable_name;`` syntax. * **Data Types** → Java has several built-in data types, including: + **Integer** → whole numbers, e.g., 1, 2, 3 + **Float** → decimal numbers, e.g., 3.14, -0.5 + **Boolean** → true or false values + **Char** → single characters, e.g., 'a', 'B' * **Operators** → Java has various operators for performing arithmetic, comparison, logical, and assignment operations. ```java // Example of declaring variables public class JavaBasics { public static void main(String[] args) { int x = 5; // declare an integer variable double y = 3.14; // declare a float variable boolean isAdmin = true; // declare a boolean variable char letter = 'A'; // declare a char variable } } ``` --- ■ **Java Control Structures** * **If-Else Statements** → used for conditional execution of code * **Loops** → used for repeated execution of code + **For Loop** → used for iterating over a sequence of values + **While Loop** → used for iterating until a condition is met ```java // Example of if-else statement public class ControlStructures { public static void main(String[] args) { int x = 5; if (x > 10) { System.out.println("x is greater than 10"); } else { System.out.println("x is less than or equal to 10"); } } } // Example of for loop public class ForLoop { public static void main(String[] args) { for (int i = 0; i < 5; i++) { System.out.println(i); } } } ``` --- ■ **Java Functions** In Java, a function is a block of code that can be executed multiple times from different parts of the program. Functions can take arguments and return values. ```java // Example of a simple function public class Functions { public static void main(String[] args) { greet("John"); } public static void greet(String name) { System.out.println("Hello, " + name + "!"); } } ``` --- ■ **Java Classes and Objects** In Java, a class is a blueprint for creating objects. An object is an instance of a class and has its own set of attributes (data) and methods (functions). ```java // Example of a simple class public class Person { private String name; private int age; public Person(String name, int age) { this.name = name; this.age = age; } public void displayInfo() { System.out.println("Name: " + name); System.out.println("Age: " + age); } } public class Main { public static void main(String[] args) { Person person = new Person("John", 30); person.displayInfo(); } } ```